

Forecasting nonstationary processes: what can be done?

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- The challenge
- Tools required
- Forecasting
- Onward journey

What do I mean by Nonstationary Time Series?

- I mean **NOT** second-order stationary.
- So, unconditional variance **changes** with time.
- Autocovariance, spectrum, etc. **change** with time.

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More interested in Variability and CI

Suppose have data $x_1, \dots, x_T$ from **stationary** series.

Want to make forecast $\hat{x}_T(h)$ made at time T for horizon h .

Want forecast using linear combination of past:

$$\begin{aligned}\hat{x}_T(1) &= \sum_{i=0}^{T-1} \varphi_i x_{T-i} \\ &= \varphi_0 x_T + \varphi_1 x_{T-1} + \varphi_2 x_{T-2} + \dots\end{aligned}$$

Note: φ sequence DOES NOT depend on time (stationary x_t)

Theory can tell us optimal least-squares forecast (Box-Jenkins).

Stationary AR(p) model:

$$X_t = \sum_{i=1}^p \varphi_i X_{t-i} + \epsilon_t.$$

For forecasting we teach UG students to:

- Decide on the most appropriate model for the data;
 - use the pacf to aid choice of p ,
 - estimate the coefficients using MLE or Yule-Walker equations:

$$\begin{bmatrix} \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_p \end{bmatrix} = \begin{bmatrix} \gamma_0 & \gamma_{-1} & \cdots \\ \gamma_1 & \gamma_0 & \cdots \\ \vdots & \vdots & \ddots \\ \gamma_{p-1} & \gamma_{p-2} & \cdots \end{bmatrix} \begin{bmatrix} \varphi_1 \\ \varphi_2 \\ \vdots \\ \varphi_p \end{bmatrix}$$

where $\gamma_{\cdot}$ is the autocovariance function for X_t .

- Forecast the data using the fitted coefficients and setting $\epsilon_t = 0$.



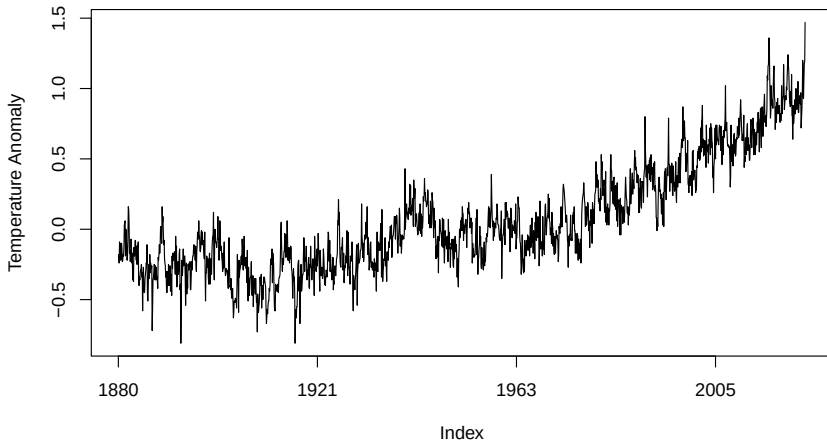
Stationary forecasting is hard to beat



Stationary forecasting is hard to beat

- Variance compensates for nonstationarity
- Can compensate with higher order models

GMST Example



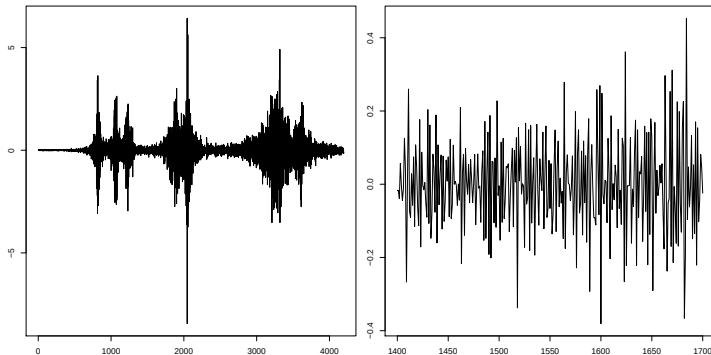


- Non-stationary model
- Approach for forecasting the nonstationary structure
- Evaluation metric (based on variability NOT mean)

Locally stationary



If you take a small enough region, it will appear stationary as the structure varies slowly over time.



Application areas include;

- medicine, finance
- environmental processes, e.g. wind speeds.

A nonstationary time series can be modelled as a trend locally stationary wavelet (T-LSW) process $\{X_{t,T}\}$, $t = 0, \dots, T - 1$, as follows:

$$X_{t,T} = \mu(t/T) + \sum_{j=-\infty}^{-1} \sum_{k \in \mathbb{Z}} w_j(k/T) \psi_{j,k-t} \xi_{j,k},$$

where

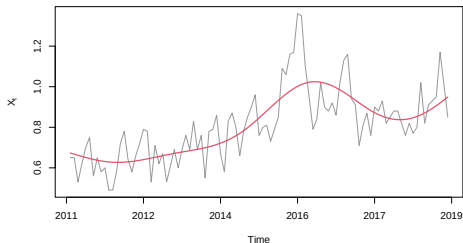
- μ is a low order polynomial mean function,
- $w_j(k/T)$ are amplitudes, the smoothness of which controls the degree of nonstationarity,
- $\{\psi_{j,k-t}\}_{j,k}$ is a set of discrete non-decimated wavelets,
- $\{\xi_{j,k}\}$ is a sequence of orthonormal random variables.

McGonigle et al. (2022) JTSA, TLSW package on CRAN

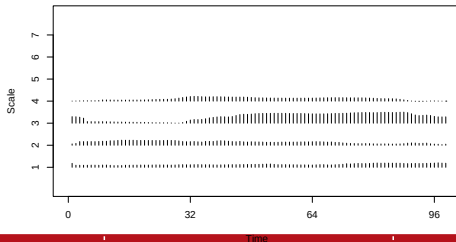
GMST Example



Trend Estimate



Spectral Estimate



The local autocovariance (LACV) is given by

$$c(z, \tau) = \sum_j S_j(z) \Psi_j(\tau),$$

- $\Psi_j(\tau)$ is the autocorrelation wavelet at scale j , lag τ
- $S_j(z)$ is the spectrum at rescaled time z

We estimate this using

$$\hat{c}(z, \tau) = \sum_{j=-J_0}^{-1} \hat{S}_j(z) \Psi_j(\tau),$$

where $J_0 = \alpha \lfloor \log_2(T) \rfloor$ for $\alpha \in (0, 1)$ is the coarsest scale used.

Given observations $x_0, \dots, x_{t-1}$, we:

- Predict $\hat{x}_t = \sum_{s=t-p}^{t-1} b_{t-1-i,T} X_s$, where
- p is the number of latest observations used for prediction and
- $\mathbf{b}$ is the solution to localized Yule-Walker equations.

$$\begin{bmatrix} c(t, 1) \\ c(t, 2) \\ \vdots \\ c(t, p) \end{bmatrix} = \begin{bmatrix} c(t-1, 0) & c(t-2, -1) & \cdots \\ c(t-1, 1) & c(t-2, 0) & \cdots \\ \vdots & \vdots & \ddots \\ c(t-1, p) & c(t-2, p-1) & \cdots \end{bmatrix} \begin{bmatrix} b_{t-1} \\ b_{t-2} \\ \vdots \\ b_{t-1-p} \end{bmatrix}$$

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These require knowledge of the covariance structure at time t which is the point we are trying to predict.

- The covariance at time t is similar to that at time $t-1$.
- We extrapolate at the smoothing step.

Thus we,

- mirror the stationary process
- build on the `ForecastLSW` approach
 - use the local partial autocorrelation function to choose p
- produce a data driven (with R package) approach for practitioners.

We define the lpacf at time t and lag τ as:

$$q(t, \tau) = \text{Corr}(X_t, X_{t-\tau} | \{X_{t-1}, \dots, X_{t-\tau+1}\})$$

$$q_T(t, \tau) = \begin{cases} \frac{c(\frac{t-1}{T}, 1)}{\sqrt{c(\frac{t}{T}, 0)c(\frac{t-1}{T}, 0)}}, & \text{for } \tau = 1 \\ \varphi_{t, \tau, \tau} \sqrt{\frac{\text{MSPE}(\hat{X}_t, X_t | X_{t-1}, \dots, X_{t-\tau+1})}{\text{MSPE}(\hat{X}_{t-\tau}, X_{t-\tau} | X_{t-1}, \dots, X_{t-\tau+1})}}, & \text{for } \tau \geq 2. \end{cases}$$

For stationary models the square root equals 1 and the $\varphi_{t, \tau, \tau}$ is the usual estimate of the pacf.

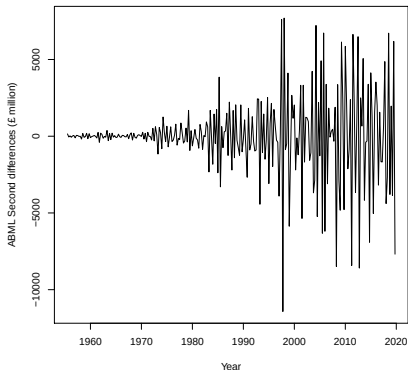
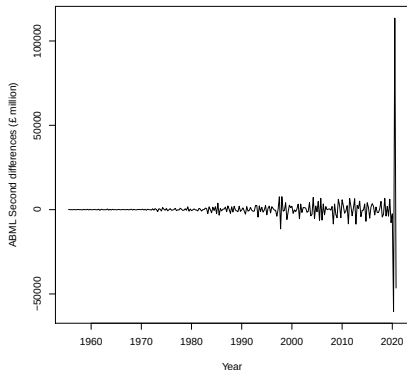
Theoretical treatment given in Killick et al. (2020) EJS, lpacf on CRAN

Range of models considered:

- TVAR(1)
- TVAR(2)
- TVAR(12)
- TVMA(1)
- TVMA(2)
- Uniformly modulated white noise
- LSW process

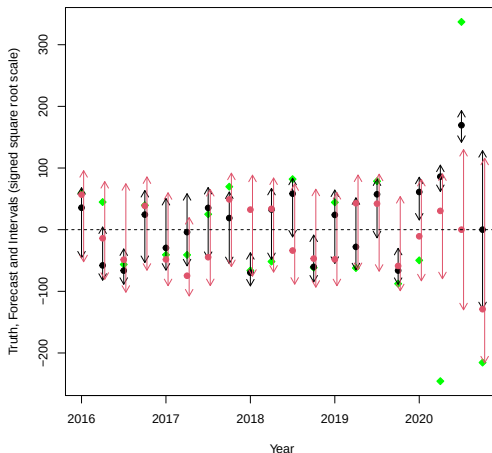
Summary

lpacf greatly improves on ARMA in 2/3 of cases
with comparable results (ratio ± 0.05) in remaining 1/3.





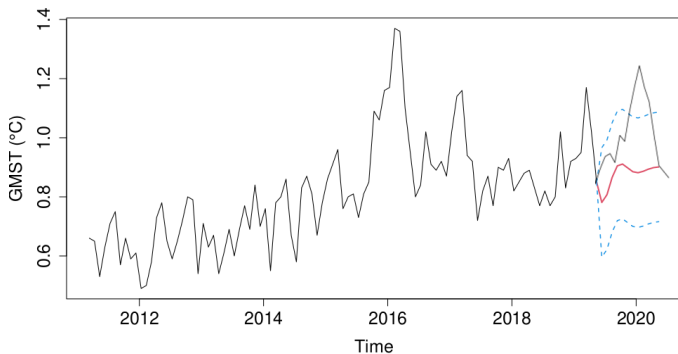
ForecastLSW package, Killick et al. JRSSC (2024+)

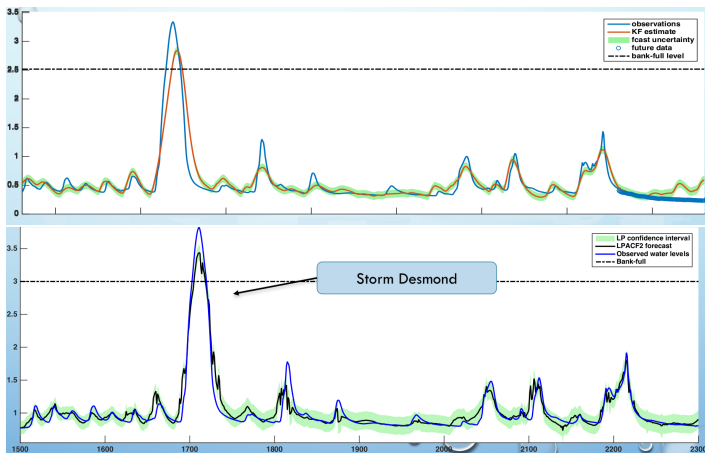




What about the mean structure?

Biggest challenge is boundary handling - Interval extension





- Non-stationary forecasting is hard
- ... but possible
- Recent advances in nonstationary methodology make forecasting possible
- Most methods are only doing point forecasts :-)
- Extensions to TLSW are showing good preliminary results
 - Now to theoretically back it up!